Swift AI Academy Pathway Engine v2 - Backend Specification

1. How to use this document

This is the build specification for the Swift AI Academy Pathway Engine. The engine recommends **courses**, not careers: it tells a learner which 1 or 2 courses to take next and why. **Build version 1 (v1) as a hard-coded, rule-based system.** Every list, mapping, weight and threshold in this document is a v1 default, loaded as seed data.

Build instructions for the AI build tool

1. Build v1 as a deterministic rules engine. The same learner input MUST always give the same persona, scores and pathway.
2. The engine runs behind the **Swift AI Academy platform's own screens** (web and mobile). There is no chatbot and no dependency on any career counselling bot.
3. The assessment is our own: the **Swift Pathway Assessment** (Sections 10 and 11). Do not call any external psychometric service. Score it inside this system.
4. Hard-code all ontologies (skills, career goals, courses, personas) and the assessment item bank as seed data in versioned JSON files. Load them through migrations.
5. Store every weight, threshold and preset in one `engine_config` table, keyed by version. Code reads values from config. No numbers are written inside the logic.
6. Use no machine learning and no LLM in scoring, persona matching or pathway building. Learners pick goals from fixed options, not free text.
7. Log every recommendation run with its inputs, intermediate scores, config version and output, so any result can be explained and replayed.
8. Keep the engine catalogue-agnostic. A new course added through the onboarding template (Section 18) MUST be scored with no code change.

Conventions

Item	Convention
Levels	One scale for skills, courses, goal areas and learners: NSQF-aligned, 0 to 8 in steps of 0.5. 0 means no measurable skill.
Scores	Goal-area and course scores run from 0 to 1.
Persona axis scores	Run from -1 to +1. 0 is the boundary between the two poles of an axis.

Assessment scales	Statements answered 1 to 5. Scale score = mean of its items (1 to 5).
IDs	Prefix + family + running number. Never reused, never renamed (SK-COMM-001, CG-02, AAI-1, PER-CFLS, LP-SD1).
MUST / SHOULD	MUST is required in v1. SHOULD is preferred and can follow later.
Level claims	Courses are described as "designed against NSQF Level X descriptors" until listed on the NQR.
Pathway	6 months with 1 course, or 12 months with 2 courses (one per 6-month term).

Reading order

Sections 2 to 4 set purpose, rules and frameworks. Sections 5 to 9 define the architecture and the reference ontologies. Sections 10 and 11 define our assessment and its full item bank. Section 12 defines the personas. Sections 13 to 15 define the engine and what learners see. Sections 16 to 19 are build references. Section 20 runs one learner end to end. Sections 21 and 22 cover quality, tuning and open items.

2. What the system does

The engine reads a learner through our own 30-minute assessment, places them on a skill level, matches them to a persona, finds the skills they are missing for the career goals they care about, and recommends a course pathway: 6 months with 1 course, or 12 months with 2 courses.

The five questions the engine answers

1. **Who is this learner?** Background, study, skill levels, interests, learning habits and goals (Sections 10 and 11).
2. **What kind of learner are they?** The closest of 16 learner personas (Section 12).
3. **Which career goals could our courses serve for them?** The top 3 of 12 career goal areas (Section 8).
4. **What skills are missing?** The gap between what those goal areas need and what the learner holds (Section 7).
5. **Which courses close that gap, in what order?** A scored, explained 6-month or 12-month pathway (Sections 13 and 14).

Inputs and outputs

Inputs	Outputs

Part A: About you (profile facts)	Learner persona with confidence
Part B: Learning profile (40 statements: interests and learning habits)	Interest profile and learning-habit scores
Part C: Skill check (AI and Communication tasks)	Skill levels on the 0 to 8 scale
Part D: Goals (goal areas, main goal, timing)	Top 3 career goal areas and the skill gap list
Skills, career goal and course ontologies (seed data)	Pathway: 1 or 2 courses, each with reasons and example roles
Learning events from the course platform	Updated pathway when something changes

Launch catalogue: 13 courses

Code	Course	Series	NSQF level (designed against)	Hours (total)	Status
AAI-1	AI First Steps: Use It and Check It	Applied AI	1	16 (proposed)	Adapt from current scope
AAI-2	Apply AI Where It Matters	Applied AI	2	16 (proposed)	Adapt from current scope
AAI-3	AI Trust, Safety and Harm	Applied AI	3	16 (proposed)	Adapt from current scope
AAI-4	Make AI Stick: Build a Tested Method	Applied AI	4	16 (proposed)	Adapt from current scope
COM-1	Everyday Workplace Communication	Communication	1 to 2	58	Framework ready
COM-2	Trade and Team Communication Skills	Communication	2.5 to 3	44	Framework ready
COM-3	Supervisory Communication Skills	Communication	3.5 to 4	57	Framework ready

COM-4	Junior Management Communication Skills	Communication	4.5 to 5	57	Framework ready
COM-5	Interpersonal Communication for Managers	Communication	5.5 to 6	64	Framework ready
COM-6	Cross-Cultural and Executive Presence Communication	Communication	6.5	46	Framework ready
COM-7	Strategic Negotiation and Crisis Leadership Communication	Communication	7	38	Framework ready
COM-8	Research, Governance and Thought-Leadership Communication	Communication	8	60	Framework ready
BOT-1	SwiftChat Bot Builder	Bot Builder	4 (proposed)	To confirm	Scope to write

The current Applied AI scope has four 16-hour units (15 training + 1 assessment), all proposed at NSQF Level 4. For the engine they become a Level 1 to 4 ladder. Hours per level are to be confirmed. BOT-1 teaches building bots on the SwiftChat developer platform; that is course content only, not a dependency of this system.

Glossary

Term	Meaning
Swift Pathway Assessment	Our own 4-part assessment: About you, Learning profile, Skill check, Goals
Learning profile	Part B: 40 statements measuring 4 interest orientations and 5 learning-habit scales
Interest orientation	One of 4 work preferences: People, Data, Ideas, Things

Career goal area	One of 12 fixed groups of work our courses can serve, such as Finance, Accounts & Banking
Example role	A job shown under a goal area to make it concrete, such as Accounts assistant
Skill	One assessable ability with an ID, measured on the 0 to 8 level scale
Learner persona	One of 16 learner types; decides how the journey is run
Term	A 6-month block. One course runs per term.
Pathway	1 term (6 months, 1 course) or 2 terms (12 months, 2 courses)
Placement	The learner's current level in a course series, from the skill check
Gap	Required level minus current level for one skill, never below 0

3. Design principles and guardrails

The engine is course-first and skills are the only bridge to careers. Every rule below is a MUST in v1.

#	Principle	What the system must do
1	Course-first	The output is always a course pathway. Careers appear only as goal areas and example roles where the course skills are used.
2	Skills are the only bridge	Never link a course directly to a goal area or role. Courses build skills; goal areas need skills. All matching runs through skill IDs.
3	Top locked, bottom grows	Skill families, the 12 goal areas, the 4 interest orientations and the 16 personas are fixed. Skills,

		example roles, courses and mappings can grow.
4	A guide, not a label	All learner-facing text uses "may suit", "likely", "could help". Never "you are" or "you should".
5	No placement promise	Never promise a job, salary or placement. No salary figures appear in v1.
6	Honest level claims	Say "designed against NSQF Level X descriptors". Never "NSQF certified" or "NCVET approved" until confirmed.
7	The learner's goal is respected	Never override the goal areas a learner picks. If they conflict with their interest profile, show both and explain.
8	Every result is explained	Every course carries reasons: skills it builds, goal areas it serves, what it unlocks.
9	Deterministic in v1	Same inputs and same config version give the same output. No randomness.
10	When unsure, support more	If a persona axis is borderline, choose the pole that gives more support and more breadth (Section 12).
11	Bad data is not used	A learning profile that fails the quality checks is dropped from scoring. The engine runs on the remaining inputs and asks for one retake.
12	Habits, not diagnoses	The learning profile measures interests and learning habits only. No clinical or personality labels are shown; scale scores go to mentors, not to learners as labels.

13	Original test content	Every assessment item is written by us. No items are copied from commercial or proprietary tests.
14	Minimum personal data	Collect only the fields in Section 11. No caste, religion, income or health data. Consent recorded under the Digital Personal Data Protection Act, 2023.
15	Catalogue-agnostic engine	A new course is scored as soon as its skills are tagged. No course name appears in engine code.

4. Frameworks used

Indian qualification frameworks set the level scale and credit. Well-established psychology frameworks shape our own assessment scales, with every item written by us. Communication and AI frameworks shape the skill list.

Framework	What it gives the engine	Where it is used
NSQF (National Skills Qualifications Framework)	Levels 1 to 8 and their descriptors	One level scale for skills, courses, goal areas and learners (Sections 7 to 9)
NCrF (National Credit Framework)	Credit from notional learning hours (30 hours = 1 credit)	Course record credits field (Section 9)
NCVET micro-credential guidelines	Stackable units; level set by entry level	Applied AI and Bot Builder design; level-claim wording
Holland RIASEC, reduced to Prediger's four work-task poles (People, Data, Ideas, Things)	A short, well-tested way to describe work interests	Part B interest orientations; goal-area fit (Sections 8, 10, 13)
Big Five: Conscientiousness	The trait most linked to finishing what you start	Basis for the Self-discipline scale (Part B)
Self-regulated learning	How learners plan, monitor and drive their own learning	Basis for the Learning independence and Learning motivation scales

Self-efficacy	A person's belief they can do a task, which predicts effort	Basis for the two Confidence scales (English, technology and AI)
Survey design practice	1 to 5 agreement scales, reverse-keyed items, attention checks, straight-line and speed checks	Part B scoring and quality flags (Section 10)
Communication programme domains (CEFR Companion Volume model)	Reception, Production, Interaction, Mediation, plus Strategic, Pragmatic, Digital	Communication skill clusters; skill check bands
LSRWVG strands	Listening, Speaking, Reading, Writing, Vocabulary, Grammar	Communication courses and skill-check items
UNESCO AI Competency Framework for Students	Candidate reference for AI competencies (alignment to be validated)	External reference on Digital & AI skills
ESCO, O*NET, SSC National Occupational Standards, NCO 2015	Public skill and occupation taxonomies	Optional crosswalk IDs on skills and example roles (not used in v1 scoring)

Two design ideas also shape the engine. Scaffolding (zone of proximal development) sets placement: the next course sits one step above the learner's current level. Knowledge graph modelling sets the data shape: nodes (learner, skill, course, goal area, role) joined by typed, weighted edges.

5. System architecture and build phases

Three clients (platform screens, Admin Console, course platform) talk to one API. Behind it sit six rule-based services and four data stores. In v1 all reference data is seed data and all logic is rules.

system architecture · 3 clients, 1 API, 6 services, 4 data stores

No chatbot sits anywhere in this picture: learners use the platform's own screens, and the assessment is scored inside the engine.

Components

#	Component	Job	v1 implementation
1	Platform screens	Run the assessment, show persona, goals, gaps and pathway	Swift AI Academy web and mobile app screens

2	Admin Console	Add courses, skills, goal mappings, assessment items; edit config with versioning	CRUD screens over the template in Section 18
3	Course platform (LMS)	Sends learning events: course started, passed, not passed, term ended	Existing course platform; calls the events API
4	Learner Service	Stores Part A and Part D answers and consent	Form schema from Section 11
5	Learning Profile Scoring	Scores Part B: interest orientations, habit scales, quality flags	Fixed scoring rules from Section 10
6	Skill Check Service	Serves and scores Part C adaptively; returns placements	Fixed item bank and cut scores
7	Persona Engine	Computes the 4 axis scores and the closest persona	Formulas and weights from engine_config
8	Recommendation Engine	Ranks goal areas, computes skill gaps, scores eligible courses	Formulas from Section 13
9	Pathway Planner	Builds 6-month or 12-month pathways; re-plans on events	Rules from Section 14
10	Data stores	Reference store, learner store, engine config, audit log	Postgres; seed JSON loaded by migration

Build phases

Phase	What it is	Data	Logic
v1: Hard-coded	Build this first. Fixed lists and rules the team can check by hand	Seed JSON for all ontologies and the item bank; skill needs	Deterministic formulas; fixed options instead of free text; no ML, no LLM

		mapped per goal area (12)	
v2: Data-driven	Same engine, richer data	Local norms for the learning profile after about 500 learners; skill needs per example role; speaking and writing tasks	Same formulas; weights tuned from v1 outcomes
v3: Learning	Engine improves from outcomes	Completion, assessment and progression data	Weight tuning and persona validation from data; optional LLM to read free-text goals

v1 is complete when a learner can finish the assessment on the platform and receive a persona and an explained pathway, and a new course can be added through the Admin Console and appear in recommendations with no code change.

6. Ontology overview

The engine uses six ontologies. Four are reference data (skills, career goals, courses, personas) and two hold learner state (learner, pathway). They connect only through typed edges.

ontology map · 6 ontologies joined by typed edges

Courses and careers never link directly. A course builds skills, a goal area needs skills, and the learner's gap is measured on one shared level scale.

The six ontologies

Ontology	Hierarchy	Fixed or grows	Defined in
Skills	Family (8) → Cluster (23) → Skill (67) → Level (0 to 8)	Families fixed; clusters rarely; skills grow	Section 7
Career goals	Goal area (12) → Example role	Goal areas fixed; roles grow	Section 8
Course	Series → Course → Module → Learning outcome	Grows with every new course	Section 9

Learner	Learner → profile facts, learning profile, skill levels, goals	One record per learner, updated by events	Sections 10 and 11
Persona	Axis (4) → Pole (2 each) → Persona (16)	Fixed	Section 12
Pathway	Pathway → Term (1 or 2) → Course slot	Created per recommendation run	Section 14

Edge types

Edge	From → To	Weight carried	Example
BELONGS_TO	Skill → Cluster → Family	none	SK-ETH-001 belongs to ETH.VER
BUILDS	Course → Skill	Output level (0 to 8)	COM-3 builds SK-COMM-007 to level 4
MAPS_TO	Learning outcome → Skill	Output level	An AAI-1 outcome maps to SK-ETH-001 at level 1
REQUIRES_COURSE	Course → Course	Hard or soft	AAI-2 requires AAI-1 (hard)
NEEDS	Goal area → Skill	Importance (core 1.0, supporting 0.5) + required level	CG-02 needs SK-COMM-007 at level 5, core
IN_GOAL	Example role → Goal area	none	Accounts assistant is in CG-02
ORIENTED_TO	Goal area → Interest orientation	Primary or secondary	CG-02 is primarily Data, secondarily People
FEEDS	Field of study → Goal area	Direct 1.0 or indirect 0.5	B.Com feeds CG-02 directly
HAS_LEVEL	Learner → Skill	Current level + source + date	Learner holds SK-COMM-009 at level 3
FITS	Learner → Goal area	Fit score 0 to 1	From interests, goals, study and market

MATCHES	Learner → Persona	Axis scores + confidence	Learner matches PER-CFLS with confidence 0.34
CONTAINS	Pathway → Term → Course	Score + reasons	Term 1 holds COM-3 at 0.90

7. Skills ontology

The skills ontology is the shared language of the engine: 8 fixed families, 23 clusters and 67 launch skills, each measured on one 0 to 8 level scale. Every current and future course and goal area is tagged against it. Before seeding, check whether the Career Counselling team already holds a skills list; if it does, map its items to these IDs instead of creating a second list.

Structure

Tier	Count at launch	Rule
Family	8	Fixed. A new family needs a design review.
Cluster	23	Added rarely, only when a whole area of skill is missing.
Skill	67	Grows. One atomic, assessable ability per record.
Level	0 to 8, steps of 0.5	Fixed scale, designed against NSQF level descriptors.

Level scale

Level	Generic descriptor (designed against NSQF descriptors)
0	No measurable skill yet
1	Does simple, familiar tasks with close guidance
2	Does routine tasks in familiar settings with some guidance
3	Does routine tasks independently using standard procedures
4	Handles varied tasks, checks own and others' work, may guide a small team

5	Takes responsibility for a work unit, adapts methods, trains others
6	Manages people and processes in complex, non-routine situations
7	Leads strategy across functions under high stakes and uncertainty
8	Creates new knowledge or governs institutions

In v1, a course builds each tagged skill to the course's own level (upper end of its NSQF band). For example COM-3 (NSQF 3.5 to 4) builds its skills to level 4.

Skill record

Field	Type	Required	Notes
skill_id	string	Yes	Format SK-<FAMILY>-<NNN>. Never reused.
name	string	Yes	Short, verb-led where possible.
definition	text	Yes	One or two sentences: what a person can do.
family_id	enum	Yes	One of the 8 families.
cluster_id	string	Yes	One of the clusters below.
level_descriptors	object	Yes	What the skill looks like at each level band the catalogue covers.
evidence_type	enum	Yes	objective_test, performance_task, portfolio, observation, self_report.
aliases	string[]	No	Other names for search. An alias never becomes a new skill.
related_skill_ids	string[]	No	Skills often learnt or used together.

transferable	boolean	Yes	True if useful across most goal areas.
external_refs	object	No	Crosswalk IDs: ESCO, O*NET, NOS, UNESCO AI CF, CEFR.
status	enum	Yes	draft, active, retired.
version	integer	Yes	Raised on any change to definition or descriptors.

Rules

1. One skill = one ID. Two names for the same ability = one skill with an alias.
2. A skill must be assessable. If no task can show it, it is a topic, not a skill.
3. Each learning outcome maps to at most 3 skills, each with an output level.
4. A skill sits in exactly one cluster and one family.
5. Skills are retired, never deleted, so old learner records stay readable.

Families and clusters

Family	ID	Clusters
Communication	COMM	REC Reception · PRO Production · INT Interaction · MED Mediation · STR Strategic · PRG Pragmatic · DIG Digital · LNG Language accuracy
Digital & AI	DAI	USE AI use and prompting · APP AI application and transfer · MTH AI methods and workflows
Safety, Ethics & Trust	ETH	VER Verification · SAF Safety, privacy and disclosure
Thinking & Problem Solving	THK	ANA Analysis and judgement · PLN Planning
People & Leadership	PPL	TEAM Team leadership · DEV Developing others · GOV Governance

Self-Management	SLF	LRN Learning and follow-through · RES Composure
Building & Technical	BLD	BOT Bot and mini-app building · INT Integration and deployment · OPS Analytics and publishing
Domain & Trade Knowledge	DOM	Reserved for future domain and partner courses

Launch skill list (67 skills)

Skill ID	Skill	Cluster	Built by (level)
SK-COMM-001	Listening for gist and detail	COMM.REC	COM-1 (2), COM-2 (3), COM-3 (4)
SK-COMM-002	Critical and comparative listening	COMM.REC	COM-1 (2), COM-3 (4), COM-5 (6)
SK-COMM-003	Reading for information (scan, skim, detail)	COMM.REC	COM-1 (2), COM-2 (3)
SK-COMM-004	Reading work documents (job cards, SOPs, reports)	COMM.REC	COM-2 (3), COM-3 (4), COM-4 (5)
SK-COMM-005	Spoken updates and self-presentation	COMM.PRO	COM-1 (2), COM-2 (3), COM-3 (4)
SK-COMM-006	Workplace writing: forms, notes, logs	COMM.PRO	COM-1 (2), COM-2 (3)
SK-COMM-007	Report and business writing	COMM.PRO	COM-3 (4), COM-4 (5), COM-7 (7)
SK-COMM-008	Presenting to groups and handling questions	COMM.PRO	COM-6 (6.5), COM-8 (8)
SK-COMM-009	Everyday workplace conversation and requests	COMM.INT	COM-1 (2), COM-2 (3)
SK-COMM-010	Giving instructions and delegating	COMM.INT	COM-3 (4), COM-4 (5)

SK-COMM-011	Giving and receiving feedback	COMM.INT	COM-3 (4), COM-4 (5), COM-5 (6)
SK-COMM-012	Customer and team interaction	COMM.INT	COM-2 (3)
SK-COMM-013	Relaying and summarising information	COMM.MED	COM-1 (2), COM-3 (4)
SK-COMM-014	Explaining complex ideas to non-experts	COMM.MED	COM-8 (8)
SK-COMM-015	Clarification and repair	COMM.STR	COM-1 (2)
SK-COMM-016	Persuasion and influence	COMM.STR	COM-5 (6), COM-6 (6.5)
SK-COMM-017	Negotiation	COMM.STR	COM-4 (5), COM-5 (6), COM-7 (7)
SK-COMM-018	Difficult conversations and conflict resolution	COMM.STR	COM-3 (4), COM-5 (6)
SK-COMM-019	Storytelling	COMM.STR	COM-6 (6.5)
SK-COMM-020	Crisis and stakeholder communication	COMM.STR	COM-7 (7)
SK-COMM-021	Register and tone	COMM.PRG	COM-1 (2), COM-2 (3)
SK-COMM-022	Cross-cultural communication	COMM.PRG	COM-6 (6.5)
SK-COMM-023	Executive presence and delivery	COMM.PRG	COM-6 (6.5)
SK-COMM-024	Email and chat messages	COMM.DIG	COM-1 (2), COM-2 (3)
SK-COMM-025	Workplace vocabulary	COMM.LNG	All COM courses (own level)
SK-COMM-026	Grammar accuracy	COMM.LNG	All COM courses (own level)
SK-COMM-027	Academic and research discourse	COMM.MED	COM-8 (8)

SK-DAI-001	AI tool setup and first use	DAI.USE	AAI-1 (1)
SK-DAI-002	Prompt framing (context, role, task, format)	DAI.USE	AAI-1 (1), AAI-2 (2)
SK-DAI-003	Output refinement (narrow, expand, change register, challenge, merge)	DAI.USE	AAI-1 (1), AAI-2 (2)
SK-DAI-004	Understanding what AI is and is not	DAI.USE	AAI-1 (1), AAI-3 (3)
SK-DAI-005	Task triage: when to use AI and when not	DAI.APP	AAI-2 (2)
SK-DAI-006	Task decomposition	DAI.APP	AAI-2 (2), AAI-4 (4)
SK-DAI-007	Applying AI in a work, study or life arena	DAI.APP	AAI-2 (2)
SK-DAI-008	Transferring an AI method to new tasks	DAI.APP	AAI-2 (2), AAI-4 (4)
SK-DAI-009	Building a repeatable AI method	DAI.MTH	AAI-4 (4)
SK-DAI-010	Testing and fault diagnosis of AI workflows	DAI.MTH	AAI-4 (4), BOT-1 (4)
SK-ETH-001	Output verification matched to stakes	ETH.VER	AAI-1 (1), AAI-2 (2), AAI-3 (3), AAI-4 (4)
SK-ETH-002	Fabrication detection	ETH.VER	AAI-1 (1), AAI-3 (3)
SK-ETH-003	Fraud and synthetic media detection	ETH.SAF	AAI-3 (3)
SK-ETH-004	Data privacy and masking	ETH.SAF	AAI-3 (3), BOT-1 (4)
SK-ETH-005	Disclosure of AI use	ETH.SAF	AAI-3 (3)
SK-ETH-006	Bias and missing-perspective awareness	ETH.VER	AAI-3 (3)

SK-ETH-007	Knowing when a qualified person must decide	ETH.SAF	AAI-3 (3)
SK-ETH-008	Hazard and safety reporting	ETH.SAF	COM-1 (2), COM-2 (3)
SK-THK-001	Numerical accuracy checking	THK.ANA	AAI-2 (2)
SK-THK-002	Source evaluation against authoritative references	THK.ANA	AAI-2 (2), AAI-3 (3), COM-8 (8)
SK-THK-003	Problem diagnosis	THK.ANA	AAI-4 (4), BOT-1 (4)
SK-THK-004	Strategic thinking and vision	THK.PLN	COM-7 (7), COM-8 (8)
SK-PPL-001	Team coordination	PPL.TEAM	COM-3 (4)
SK-PPL-002	Motivating and guiding a team	PPL.TEAM	COM-4 (5), COM-5 (6)
SK-PPL-003	Training others	PPL.DEV	COM-4 (5)
SK-PPL-004	Mentoring and coaching	PPL.DEV	COM-5 (6)
SK-PPL-005	Managing upward and cross-functional work	PPL.TEAM	COM-4 (5), COM-5 (6)
SK-PPL-006	Handling grievances	PPL.TEAM	COM-4 (5)
SK-PPL-007	Governance and board-level judgement	PPL.GOV	COM-8 (8)
SK-SLF-001	Planning and follow-through	SLF.LRN	AAI-4 (4)
SK-SLF-002	Self-directed learning and transfer	SLF.LRN	AAI-2 (2)
SK-SLF-003	Composure under pressure	SLF.RES	COM-6 (6.5), COM-7 (7)
SK-BLD-001	Conversational flow design	BLD.BOT	BOT-1 (4)

SK-BLD-002	AI-assisted prototyping of bots and mini-apps	BLD.BOT	BOT-1 (4)
SK-BLD-003	Deploying and embedding a bot	BLD.INT	BOT-1 (4)
SK-BLD-004	Login, single sign-on and access control basics	BLD.INT	BOT-1 (4)
SK-BLD-005	Connecting a bot to data through APIs	BLD.INT	BOT-1 (4)
SK-BLD-006	Usage analytics and iteration	BLD.OPS	BOT-1 (4)
SK-BLD-007	Publishing and listing a bot	BLD.OPS	BOT-1 (4)
SK-BLD-008	Testing and debugging a bot	BLD.OPS	BOT-1 (4)

Skill names and course mappings are the v1 seed. Course teams confirm each mapping against their learning outcomes before launch (Section 22).

8. Career goal ontology

Careers enter the engine as 12 fixed career goal areas that we define for our course catalogue. Each goal area lists example roles, the interest orientation it suits, the fields of study that lead to it, a market outlook, and the skills it needs. This replaces any dependency on an external career knowledge base.

Goal area record

Field	Type	Notes
goal_id	string	CG-01 to CG-12. Fixed.
name	string	Learner-facing name
definition	text	One or two plain sentences: what this work is
primary_orientation, secondary_orientation	enum	People, Data, Ideas or Things
market_outlook	integer 1 to 5	Set by the research team from labour-market sources; reviewed yearly

example_roles	list	3 to 6 job titles; grows over time; optional NCO 2015 code
kb_track_ids	list	Optional crosswalk to the Career Knowledge Base; not used in v1
status	enum	active, retired

The 12 goal areas

Market outlook values are illustrative until the research team confirms them.

ID	Goal area	Example roles	Orientation (primary, secondary)	Outlook
CG-01	Office & Administration	Office assistant, data entry operator, front office executive, admin executive	Data, People	3
CG-02	Finance, Accounts & Banking	Accounts assistant, billing executive, bank customer service officer, banking correspondent	Data, People	4
CG-03	Sales, Marketing & Customer Service	Sales associate, customer care executive, telecaller, digital marketing assistant	People, Ideas	4
CG-04	Hospitality, Retail & Tourism	Front desk associate, retail store associate, travel desk executive	People, Things	3

CG-05	Teaching & Training	Teaching assistant, trainer assistant, community trainer	People, Ideas	3
CG-06	Healthcare Support	Hospital front office executive, patient care coordinator, medical records assistant	People, Data	4
CG-07	Skilled Trades & Maintenance	Electrician, fitter, maintenance technician, service technician	Things, Data	4
CG-08	Manufacturing & Supervision	Line supervisor, quality inspector, shift in-charge	Things, People	3
CG-09	IT Support & Digital Operations	IT helpdesk executive, computer operator, digital service centre operator	Things, Data	4
CG-10	Software, Bots & Apps	Junior developer, chatbot builder, no-code app builder, QA tester	Ideas, Data	5
CG-11	Data & Analytics	Data analyst trainee, MIS executive, reporting assistant	Data, Ideas	5
CG-12	Media, Content & Communication	Content writer, social media assistant, video editing assistant	Ideas, People	3

Starting a business is treated as a goal type (Part D), not a goal area, because it can sit inside any of the 12.

Skill needs per goal area (seed)

Each goal area lists 4 to 8 skills. Core counts 1.0 and supporting 0.5 in the engine. Numbers in brackets are required levels. To be confirmed by course and research teams.

Goal area	Core skills (level)	Supporting skills (level)
CG-01	COMM-024 Email and chat (4); COMM-007 Report writing (4); COMM-004 Reading work documents (4); DAI-002 Prompt framing (3); ETH-004 Data privacy (4)	COMM-009 Workplace conversation (4); THK-001 Numerical accuracy (3)
CG-02	COMM-007 Report writing (5); COMM-004 Reading work documents (4); ETH-001 Output verification (4); THK-001 Numerical accuracy (4)	COMM-009 Workplace conversation (4); DAI-002 Prompt framing (3)
CG-03	COMM-009 Workplace conversation (4); COMM-016 Persuasion (5); COMM-024 Email and chat (4)	DAI-002 Prompt framing (3); COMM-019 Storytelling (5)
CG-04	COMM-009 Workplace conversation (3); COMM-012 Customer interaction (3); COMM-021 Register and tone (3); COMM-015 Clarification (3)	COMM-022 Cross-cultural (4); DAI-002 Prompt framing (2)
CG-05	COMM-013 Summarising (4); COMM-011 Feedback (4); PPL-003 Training others (5); COMM-008 Presenting (5)	DAI-002 Prompt framing (3); ETH-001 Output verification (3)
CG-06	COMM-009 Workplace conversation (4); COMM-004 Reading work documents (4); ETH-004 Data privacy (4); COMM-006 Workplace writing (3)	COMM-018 Difficult conversations (4); ETH-007 When a qualified person decides (3)
CG-07	COMM-001 Listening (3); COMM-004 Reading work documents (3); ETH-008 Hazard reporting (3)	COMM-006 Workplace writing (3); DAI-002 Prompt framing (2); ETH-001 Output verification (2)

CG-08	COMM-010 Giving instructions (4); PPL-001 Team coordination (4); ETH-008 Hazard reporting (3); COMM-007 Report writing (4)	COMM-018 Difficult conversations (4); DAI-002 Prompt framing (2)
CG-09	THK-003 Problem diagnosis (4); COMM-009 Workplace conversation (4); DAI-005 Task triage (3); ETH-004 Data privacy (4)	COMM-024 Email and chat (4); DAI-010 Workflow testing (3)
CG-10	BLD-001 Flow design (4); BLD-002 AI-assisted prototyping (4); BLD-005 Data through APIs (4); DAI-010 Workflow testing (4); THK-003 Problem diagnosis (4)	ETH-004 Data privacy (4); COMM-007 Report writing (4)
CG-11	THK-001 Numerical accuracy (4); ETH-001 Output verification (4); DAI-002 Prompt framing (3); DAI-006 Task decomposition (3)	COMM-007 Report writing (4)
CG-12	COMM-019 Storytelling (5); COMM-007 Report writing (4); DAI-003 Output refinement (3); ETH-005 Disclosure of AI use (3)	ETH-002 Fabrication detection (3); COMM-021 Register and tone (4)

Some required levels sit above what any launch course builds (for example numerical accuracy to 4). The engine reports these as "build elsewhere" (through the learner's degree, trade or other certificates).

Field of study to goal area (FEEDS)

Field of study or trade	Direct (1.0)	Indirect (0.5)
B.Com	CG-02	CG-01, CG-11
BCA / B.Sc Computer Science	CG-10	CG-09, CG-11
BA (any)	none	CG-05, CG-12, CG-01
B.Ed	CG-05	none
B.Sc Nursing, GNM, paramedical	CG-06	none

Hotel management	CG-04	CG-03
ITI Electrician, Fitter, Mechanic	CG-07	CG-08, CG-09
ITI COPA	CG-09	CG-01

The full table covers every stream and ITI trade at partner institutions (open item, Section 22).

9. Course ontology

A course is a tagged bundle of skills with an entry rule. The engine never reads course content; it reads the course's skills, output levels, prerequisites and eligibility. Hierarchy: Series → Course → Module → Learning outcome → Skill.

Course structure

Level	Key fields	Example
Series	series_id, name, level ladder	COM: Communication, 8 levels
Course	course_id, title, NSQF level, hours, entry level, prerequisites, skills built	COM-3 Supervisory Communication Skills
Module	module_id, title, hours (theory, practical, assessment), assessment weight	COM-3 M2 Listening to the Shop Floor and Reporting Upward
Learning outcome	lo_id, text, type (theory or practical), strand, domain, mapped skills	"Give a short spoken update to the person in charge" → SK-COMM-005 at level 4

Entry rules and prerequisites

Placement is the learner's level in that series from the skill check (Section 10). A learner meets the entry rule if they completed the previous course OR their placement is at or above the entry level.

Course	NSQF (designed against)	Training hours (theory + practical)	Entry level	Prerequisite	Other eligibility (proposed)
AAI-1	1	15	0	None	Age 17 or over
AAI-2	2	15	1	AAI-1 (hard)	None
AAI-3	3	15	2	AAI-2 (hard)	None

AAI-4	4	15	3	AAI-3 (hard)	None
COM-1	1 to 2	50	0	None	None
COM-2	2.5 to 3	38	2	COM-1 (hard)	None
COM-3	3.5 to 4	50	3	COM-2 (hard)	None
COM-4	4.5 to 5	50	4	COM-3 (hard)	None
COM-5	5.5 to 6	56	5	COM-4 (hard)	Graduate, or 2 years of work experience
COM-6	6.5	40	6	COM-5 (hard)	Managerial work experience
COM-7	7	32	6.5	COM-6 (hard)	Senior managerial work experience
COM-8	8	53	7	COM-7 (hard)	Doctoral research, or board or senior leadership role
BOT-1	4 (proposed)	To confirm	AI placement 1	AAI-1 (hard); COM-2 (soft)	Class 12 passed

A hard prerequisite blocks the course. A soft prerequisite lowers its score (× 0.8) but does not block it. COM-6 to COM-8 will rarely be eligible for learners aged 17 to 30; they stay for completeness and working professionals.

Applied AI ladder: Levels 1 to 4

Verification runs through every level, so a learner who stops after any level can still check AI output.

Course	Terminal outcome	Teaching sections	Gate (must pass)	Rulebook sections closed
AAI-1 Use It and Check It	Use an approved AI tool on a routine task and check the result before using it	First contact · Framing and refining · Check before you use	Find planted fabrications in a timed task	R1 Tools I use · R3 How I check

AAI-2 Apply It Where It Matters	Apply AI to a chosen arena (trade, study or life) and transfer the method to unfamiliar tasks	Choose your arena · Deep practice · Transfer to two others	Complete an untaught arena task with no help	To assign
AAI-3 Trust, Safety and Harm	Detect error, fraud and manipulation in AI content and use AI safely	Fabrication and verification depth · Fraud and synthetic media · Privacy, disclosure and ethics	Classify real and synthetic media; mask and disclose	R2 What I never type in · R3 deepened · R4 What I disclose
AAI-4 Make It Stick	Build, test and demonstrate a repeatable AI method for a weekly task	Build the method · Test and harden it · Demonstrate and commit	Run the method on three inputs, including one awkward input	R5 My method · R6 My 90-day plan

Communication ladder: Levels 1 to 8

Every Communication course covers all six LSRWVG strands and closes with a 40% rubric-based capstone.

Course	Learner role it is pitched at	Modules
COM-1	Helper or ground-level worker to worker or assistant	Greetings, names and simple instructions · Safety words, requests and reporting a problem · Everyday conversations and multi-step instructions · Reading, forms and fuller reporting · A full day at work (capstone)
COM-2	Entry-skilled employee or ITI trainee	Shop floor instructions and safety · Customer and team interaction basics · Job cards, SOPs and simple logs (capstone)

COM-3	Senior skilled technician or junior supervisor	Clear instructions and team coordination · Listening to the shop floor and reporting upward · Workplace conflict and feedback · Supervisory reports and handover notes (capstone)
COM-4	Junior manager or technical supervisor	Delegating and performance feedback · Motivating, guiding and training a team · Managing upward and cross-functional coordination · Grievances, negotiation and management reports (capstone)
COM-5	Manager or senior manager	Listening to respond · Motivating teams through language · Difficult conversations and conflict · Persuasion, negotiation and influence · Mentoring, coaching and cross-team work (capstone)
COM-6	Senior manager or director	Cross-cultural communication · Executive presence and delivery · Strategic storytelling and persuasion (capstone)
COM-7	Director or senior executive	International negotiation · Crisis and stakeholder communication · Executive writing for boards and the public (capstone)
COM-8	Doctoral scholar, board member or chairperson	Defending original work · Thought leadership · Literature synthesis and peer review · Governance and strategic vision (capstone)

Bot Builder (proposed outline)

BOT-1 teaches learners to design, build and publish a bot using AI-assisted coding on the SwiftChat developer platform. Proposed until the course scope is written.

Module	What the learner does	Main skills
M1 Design a useful bot	Pick a real problem and user; draw the conversation flow	SK-BLD-001
M2 Build it with AI help	Scaffold from a template; build with AI-assisted coding, SDK and CLI	SK-BLD-002, SK-DAI-010
M3 Connect data and login	Call an API; add login, single sign-on or third-party sign-in	SK-BLD-004, SK-BLD-005, SK-ETH-004
M4 Deploy and embed	Deploy as an app, standalone, or embedded in a website or LMS	SK-BLD-003
M5 Test, publish and improve (capstone)	Debug in the sandbox, publish a listing, read usage analytics, improve	SK-BLD-006, SK-BLD-007, SK-BLD-008, SK-THK-003

10. Swift Pathway Assessment: design

The Swift Pathway Assessment is our own 4-part, roughly 30-minute assessment, built for one job: choosing courses. Every answer is used twice: once as a dated Day-0 fact (the baseline) and once as an input to the learner's profile and persona (the psychometric use).

The four parts

Part	What it is	Items	Time	Produces
A. About you	Profile facts, all multiple choice	14 questions	5 min	Education, field, experience, device, time available
B. Learning profile	Statements answered 1 to 5	40 statements	8 to 10 min	4 interest orientations, 5 learning-habit scales, quality flags
C. Skill check	Short auto-scored tasks	8 AI items + 6 to 12 Communication items (adaptive)	10 to 12 min	AI placement (0 to 4), Communication placement (0 to 8), diligence flag

D. Goals	Goal-area cards and short choices	5 questions	3 min	Chosen goal areas, main goal, earning need, mobility

What Part B measures

Scale	Items	What it tells us	Used for
Interest: People	4	Enjoys helping, explaining, working with others	Goal-area fit; Direction axis
Interest: Data	4	Enjoys records, numbers, procedures, organising	Goal-area fit; Direction axis
Interest: Ideas	4	Enjoys creating, designing, researching	Goal-area fit; Direction axis
Interest: Things	4	Enjoys tools, machines, devices, hands-on work	Goal-area fit; Direction axis
Self-discipline	5	Plans and finishes what they start	Learning Drive axis
Learning independence	4	Comfortable learning alone, finds answers	Learning Drive axis; support mode
Learning motivation	4	Wants to keep improving, learns beyond requirements	Learning Drive axis
Goal clarity	3	Knows what work they want and what it needs	Direction axis
Confidence: English	3	Belief in own English communication	Learning Drive axis; confidence check for mentors
Confidence: Technology and AI	3	Belief in own ability with apps and AI tools	Learning Drive axis; confidence check for mentors
Quality items	2	Attention checks	Quality flag

How one assessment is both baseline and psychometric

Part	As a baseline (Day-0 record, re-measured later)	As a psychometric and persona input
A	Education, experience and access on Day 0	Starting Point, Goal Horizon and Learning Drive axes; Field term in goal ranking
B	Interests and habits on Day 0; habit scales re-measured each term to show growth	Interest orientations drive goal-area fit; habit scales drive Learning Drive; goal clarity and interest focus drive Direction
C	Skill levels on Day 0; re-measured at each term end to show skill growth	Starting Point axis; behaviour (finishing in one sitting at normal pace) drives Learning Drive
D	Stated goals on Day 0; re-asked each term	Direction and Goal Horizon axes; Aspiration term in goal ranking

Re-measurement schedule. Day 0: full assessment. End of every 6-month term: Part C for the series studied, Part B habit, clarity and confidence scales (22 items), Part D. Interest items: only after 12 months, because interests change slowly.

Scoring Part B

1. Each statement is answered 1 to 5. Interest items: Not at all (1) to Very much (5). All others: Strongly disagree (1) to Strongly agree (5).
2. Reverse-keyed items (marked R in Section 11) are flipped: score = 6 − answer.
3. Scale score = the mean of its items, from 1 to 5.
4. Engine value for habit scales = (scale score − 3) ÷ 2, giving −1 to +1.
5. Engine value for interests = (scale score − 1) ÷ 4, giving 0 to 1.
6. Interest profile: rank the 4 orientations. Top orientation = highest; second = next. Interest focus = top score − second score.
7. v1 uses these raw scores. There is no norm group yet. After about 500 learners, v2 builds local norms (percentiles and stens) per learner group.

Confidence check (for mentors only, not scored). Compare Confidence: English with Communication placement, and Confidence: Technology and AI with AI placement. Label each Over-confident, Matched or Under-confident by fixed bands (open item: set bands after the pilot). Mentors use it to start conversations.

Quality checks

Check	Rule	If it fails

Attention item LP-Q1	Answer must be 4 (Agree)	Flag
Attention item LP-Q2	Answer must be 1 or 2	Flag
Straight-lining	Same answer on 36 or more of the 40 Part B items	Flag
Speeding	Part B finished in under 120 seconds	Flag

Any flag: Part B is stored but not used in scoring; the engine drops every Part B parameter and rescales the rest (Section 12). The learner is invited to retake Part B once, the next day or later.

Skill check (Part C)

AI ladder (8 items, 2 per level). Start at Level 1. Both items correct → move up one level. Otherwise stop. Placement = highest level where both items were correct; Level 1 failed → 0.

Communication ladder (3 items per band, adaptive). Each band matches one Communication course level.

Band	Matches course	Level if passed
1	COM-1	2
2	COM-2	3
3	COM-3	4
4	COM-4	5
5 to 8	COM-5 to COM-8	6, 6.5, 7, 8

1. Start band from education stage: Class 10 or below → 1; Class 12 or ITI → 2; UG → 3; graduate or above → 4.
2. Each band has 3 items: one listening (short audio), one reading, one vocabulary or grammar.
3. 2 or 3 correct → passed → move up one band. Fewer → if this was the start band, move down once; otherwise stop.
4. Placement = level of the highest band passed. Band 1 failed → 0.

Diligence flag. Yes if all four parts were finished in one sitting and total time was between 0.5 and 2 times the expected time (proposed).

Setting starting skill levels

Skill family	Starting level
Communication, and SK-ETH-008	Communication placement
Digital & AI, and Safety, Ethics & Trust (except SK-ETH-008)	AI placement

People & Leadership	From role level in Part A: none 0, worker 2, supervisor 4, manager 5
Thinking, Self-Management, Building	0, unless a completed course or verified certificate says otherwise

When sources disagree, trust: course assessment passed > skill check > verified certificate > self-report. Self-report never raises a level above 1.

11. Assessment item bank

This is the v1 content of the Swift Pathway Assessment. All items are original. Every item ships in English and Hindi; Hindi versions are checked by back-translation before launch. Items live in assessment_items.json with IDs, options, keys and scoring rules.

Part A: About you (14 questions)

Consent screens come first (Section 17). All questions are single choice unless marked.

ID	Question	Options
A1	Your full name	Text (for records and certificates only)
A2	Your date of birth	Date. Under 17: stop. Age 17: guardian consent needed
A3	Gender (optional)	Woman, man, prefer to describe, prefer not to say. Reporting only; never scored
A4	Your state and district	Lists
A5	Which language do you want to learn in?	English, Hindi, other
A6	Where are you now?	ITI, college, university, school, not studying, working
A7	What are you studying, or what did you last complete?	Class 10 or below, Class 12, ITI year 1, ITI year 2, UG year 1, 2, 3, 4, graduate, postgraduate, doctoral
A8	How much time is left in your current studies?	2 or more years, about 1 year, final year, finished or not studying
A9	Your field of study or trade	List from the FEEDS table

A10	Certificates you already hold (multi-select)	Course list, plus "none"
A11	Your work experience	None, under 1 year, 1 to 2 years, over 2 years; plus role level: worker, supervisor, manager
A12	Which devices can you use for learning?	Own smartphone and a laptop or lab; own smartphone only; shared phone
A13	How is your internet?	Reliable, patchy, mostly offline
A14	How many hours a week can you give to a course?	Under 3, 3 to 5, 6 to 8, over 8

Part B: Learning profile (40 statements)

Shown one per screen in a fixed, interleaved order stored in the item bank, so items of one scale never sit together. Key: + scored as answered; R reverse-scored.

Interests: "How much would you enjoy..." (Not at all 1 to Very much 5)

ID	Statement	Scale
LP-P1	Helping a customer solve a problem	People
LP-P2	Explaining something to a group of people	People
LP-P3	Working in a team to plan an event	People
LP-P4	Guiding a new classmate or colleague	People
LP-D1	Keeping accurate records or accounts	Data
LP-D2	Checking numbers in a report for mistakes	Data
LP-D3	Organising information in a spreadsheet	Data
LP-D4	Following a clear procedure step by step	Data

LP-I1	Thinking up a new way to solve a problem	Ideas
LP-I2	Writing a story, post or script	Ideas
LP-I3	Designing a poster, app screen or product	Ideas
LP-I4	Researching a topic and making sense of it	Ideas
LP-T1	Repairing or setting up a machine or device	Things
LP-T2	Building or assembling something with tools	Things
LP-T3	Setting up a phone, computer or network	Things
LP-T4	Working on a shop floor, site or outdoors	Things

Learning habits, goals and confidence (Strongly disagree 1 to Strongly agree 5)

ID	Statement	Scale	Key
LP-SD1	When I start something, I finish it.	Self-discipline	+
LP-SD2	I plan my week before it starts.	Self-discipline	+
LP-SD3	I keep going even when a task becomes boring.	Self-discipline	+
LP-SD4	I often leave work until the last minute.	Self-discipline	R
LP-SD5	I keep my study or work material organised.	Self-discipline	+
LP-LI1	I like to learn on my own, at my own pace.	Learning independence	+
LP-LI2	When I get stuck, I can usually find the answer myself.	Learning independence	+

LP-LI3	I need someone to check on me to keep learning.	Learning independence	R
LP-LI4	I can learn from videos and online material without a teacher.	Learning independence	+
LP-LM1	I enjoy learning new skills even when nobody asks me to.	Learning motivation	+
LP-LM2	I want to keep getting better at what I do.	Learning motivation	+
LP-LM3	I only study when there is an exam.	Learning motivation	R
LP-LM4	I look for chances to try new things.	Learning motivation	+
LP-GC1	I know what kind of work I want to do.	Goal clarity	+
LP-GC2	I know which skills I need for the work I want.	Goal clarity	+
LP-GC3	I keep changing my mind about what I want to do.	Goal clarity	R
LP-CE1	I can explain my ideas clearly in English.	Confidence: English	+
LP-CE2	I can write a short email or message in English without help.	Confidence: English	+
LP-CE3	I feel comfortable speaking English at college or work.	Confidence: English	+
LP-CT1	I can learn a new app or tool quickly.	Confidence: Technology and AI	+
LP-CT2	I am comfortable using AI tools such as chat assistants.	Confidence: Technology and AI	+

LP-CT3	I can fix basic problems with my phone or computer.	Confidence: Technology and AI	+
LP-Q1	To show you are reading carefully, choose "Agree" for this statement.	Quality	Must be 4
LP-Q2	I have never used a mobile phone.	Quality	Must be 1 or 2

Part C: AI skill check (8 items)

Each item has 4 options; the correct option is marked ✓.

ID	Level	Item	Options
SC-AI-1a	1	An AI assistant confidently tells you the name of your state's Chief Minister. What should you do before using this answer?	A. Use it, AI is always right · B. Check it with an official source ✓ · C. Ask the same AI again · D. Share it with friends
SC-AI-1b	1	Which request will get the most useful answer?	A. "Leave letter" · B. "Write a short, polite leave letter to my principal for 2 days of sick leave, in simple English" ✓ · C. "Help me" · D. "Letter please"
SC-AI-2a	2	An AI adds 4,000 + 3,500 + 4,000 and says the total is 12,500. What is true?	A. The total is right · B. The total should be 11,500 ✓ · C. AI never makes number mistakes · D. You cannot check this

SC-AI-2b	2	Which task should you NOT hand to an AI tool to decide?	A. A first draft of a notice · B. The medicine dose for a patient ✓ · C. Ideas for a poster · D. Rewriting text in simpler words
SC-AI-3a	3	You get a voice note that sounds like your manager, asking you to transfer money urgently to a new account. What do you do?	A. Transfer it quickly · B. Reply to the voice note to confirm · C. Call your manager on their known number before doing anything ✓ · D. Forward it to a colleague
SC-AI-3b	3	Before pasting a customer complaint into an AI tool to draft a reply, what must you remove?	A. Nothing · B. The complaint itself · C. The customer's name, phone number and account number ✓ · D. The date
SC-AI-4a	4	You use the same AI prompt every week to summarise a report. One week the summary misses a key number. What is the best fix?	A. Stop using AI · B. Add a step to your method: check key numbers against the source every time ✓ · C. Use a longer prompt · D. Ask a friend
SC-AI-4b	4	You want to know if your weekly AI method is reliable. Which is the best test?	A. Run it once on this week's report · B. Run it on three past reports, including one unusual one, and check each output ✓ · C. Ask the AI if it is reliable · D. Use it for a month and see

Part C: Communication skill check (3 items per band)

Each band has one listening item (20 to 40 second audio), one reading item and one vocabulary or grammar item. One sample per band is shown; the content team writes the rest to the same pattern.

Band	Level	Sample item	Correct answer
1	2	Notice: "Canteen closed today. Lunch in Room 4." Where is lunch today?	Room 4
2	3	Job card: "Check oil level before starting the machine. Report leaks to the supervisor." What must you do first?	Check the oil level
3	4	Which handover note is clearest? (4 notes shown)	"Machine 3 stopped at 4 pm: belt fault. New belt arrives 10 am tomorrow. Do not restart until it is fitted."
4	5	A team member missed a deadline. Which is the best feedback? (4 options)	Private, specific about what happened, focused on the work, ends with an agreed plan
5	6	A colleague resists a new process. Which reply is most persuasive? (4 options)	Names their concern, gives evidence, offers a small trial
6	6.5	Which email opening suits a formal international client? (4 options)	Formal greeting, clear purpose, no idioms or slang
7	7	Which is the best first public statement in a product-safety problem? (4 options)	Acknowledges the problem, states actions taken, gives time of next update, blames no one
8	8	Which sentence reports this study correctly? "200	"The study found a link between X and Y in 200 learners; it does not show that X causes Y."

		learners; X and Y rose together." (4 options)	

Part D: Goals (5 questions)

ID	Question	Options
D1	Which kinds of work interest you? Pick up to 2.	The 12 goal-area cards, each with a one-line description and 3 example roles; or "Not sure yet"
D2	What is your main goal right now?	Job within 6 months; job in 1 to 2 years; higher studies; grow in my current job; start a business; not sure
D3	How soon do you need to start earning?	Within 6 months; 6 to 12 months; 1 to 2 years; not a priority now
D4	Where would you like to work?	Within my district; within my state; anywhere in India; abroad
D5	Anything else you want us to know? (optional)	Free text; stored for mentors; not scored in v1

12. Learner personas

Every learner is matched to the closest of 16 fixed personas, built from 4 axes with 2 poles each. The persona decides how the journey runs: pathway length, focus or breadth, starting depth and support level. The career goal areas decide where the skills point.

Two lenses, kept separate

Lens	Question it answers	Source
Goal-area fit	Which kinds of work could our courses serve for me?	Interests (Part B), chosen goals (Part D), field of study (Part A), market outlook
Learner persona	How should my learning journey be run?	All four parts of the assessment

Each axis passes four tests: it can be measured by our assessment, it changes a decision the engine makes, it is mostly independent of the others, and it can change as the learner grows.

The 4 axes

Axis	Negative pole (−1)	Positive pole (+1)	What it changes
1. Direction	E: Exploring	C: Clear	Weight on chosen goals vs breadth
2. Starting Point	F: Foundation	R: Ready	Entry level
3. Goal Horizon	N: Near-term	L: Long-term	Pathway length: 6 or 12 months
4. Learning Drive	M: Mentored	S: Self-driven	Support mode: check-ins, mentor priority, milestone size

Persona code = the four letters in axis order, for example PER-CFLS.

Parameters behind each axis

Each parameter becomes a value from −1 to +1, is multiplied by its weight, and the results are added. Weights in each axis sum to 1.0. Habit scale values come from Section 10: (scale score − 3) ÷ 2.

Axis 1. Direction (positive = Clear)

Parameter	Source	Conversion to −1 … +1	Weight
Goal clarity	Part B scale	(score − 3) ÷ 2	0.35
Named goal areas	D1	one area +1; two areas +0.5; not sure −1	0.30
Interest focus	Part B interests	top minus second: 1.0 or more +1; 0.5 to under 1.0 +0.5; 0.25 to under 0.5 −0.5; under 0.25 −1	0.20
Goal-interest agreement	D1 vs Part B	first chosen area's primary orientation is the learner's top interest +1; their second interest 0; otherwise −1; none chosen 0	0.15

Axis 2. Starting Point (positive = Ready)

Parameter	Source	Conversion to −1 … +1	Weight
AI placement	Part C	(placement − 2) ÷ 2	0.35
Communication placement	Part C	(placement − 3) ÷ 3, capped at +1	0.35
Education and experience	A7, A11	Class 10 or below −1; Class 12 or ITI year 1 −0.5; ITI year 2 or UG year 1 to 2 0; UG year 3 to 4 or graduate +0.5; postgraduate, doctoral, or over 2 years of work +1	0.20
Device access	A12	own smartphone and laptop or lab +1; own smartphone only 0; shared phone −1	0.10

Axis 3. Goal Horizon (positive = Long-term)

Parameter	Source	Conversion to −1 … +1	Weight
Main goal	D2	job within 6 months −1; grow in current job −0.5; job in 1 to 2 years 0; not sure 0; start a business +0.5; higher studies +1	0.50
Earning need	D3	within 6 months −1; 6 to 12 months −0.3; 1 to 2 years +0.3; not a priority now +1	0.30
Time left in studies	A8	2 or more years +1; about 1 year 0; final year −0.5; finished or not studying −1	0.20

Axis 4. Learning Drive (positive = Self-driven)

Parameter	Source	Conversion to −1 … +1	Weight

Self-discipline	Part B scale	(score − 3) ÷ 2	0.30
Weekly hours	A14	under 3 −1; 3 to 5 −0.3; 6 to 8 +0.3; over 8 +1	0.20
Learning motivation	Part B scale	(score − 3) ÷ 2	0.15
Learning independence	Part B scale	(score − 3) ÷ 2	0.15
Diligence flag	Part C	yes +1; no −1	0.10
Confidence	Part B, mean of both confidence scales	(score − 3) ÷ 2	0.10

If Part B is flagged, drop every Part B parameter and divide the remaining weights by their sum so they again add up to 1.0.

Matching to the closest persona

1. **Compute** the 4 axis scores. The learner is a point in 4-dimensional space, each axis −1 to +1.
2. **Nearest persona.** Each persona has a centre at ±0.5 on every axis. The learner gets the persona whose centre is nearest (Euclidean distance). In v1 this equals the sign of each axis; it is built as nearest-centre so v2 can move centres using real data with no API change.
3. **Borderline axes.** An axis with absolute score under 0.15 is borderline. Ask its tie-breaker (below), at most 2, closest-to-zero first. Add the answer as an extra parameter worth 0.25, rescale weights to sum to 1.0, recompute.
4. **Still unsure** → take the more supportive pole: Exploring, Foundation, Near-term (shorter commitment, re-planned sooner), Mentored.
5. **Confidence** = average of the 4 absolute axis scores. Under 0.25 Low; 0.25 to 0.5 Medium; above 0.5 High.
6. **Second choice** = flip the axis with the smallest absolute score. Shown to mentors.

Axis	Tie-breaker question	Scoring
Direction	"If you had to pick one of these three goal areas today, which would it be?" (top 3 shown)	picks one +1; none fit −1
Starting Point	One extra skill-check item at the borderline level	correct +1; wrong −1
Goal Horizon	"A year from now, what matters more: already earning, or being better qualified?"	earning −1; qualified +1

Learning Drive	"What helps you more: a clear weekly plan with reminders, or freedom to study at your own pace?"	plan −1; freedom +1

The 16 personas

Code	Persona	In one line	Default pathway	Preset	Support
PER-CFNS	The Sprinter	Knows the goal, starts from basics, needs results soon, works alone	6 months	Focus	Light
PER-CFNM	The Apprentice	Knows the goal, starts from basics, needs results soon, learns best with guidance	6 months	Focus	Guided
PER-CFLS	The Climber	Knows the goal, starts from basics, builds over time, works alone	12 months	Focus	Light
PER-CFLM	The Steady Builder	Knows the goal, starts from basics, builds over time, learns best with guidance	6 months first, then extend	Focus	Guided
PER-CRNS	The Launcher	Knows the goal, ready for higher levels, needs results	6 months	Focus	Light

		soon, works alone			
PER-CRNM	The Candidate	Knows the goal, ready for higher levels, needs results soon, learns best with guidance	6 months	Focus	Guided
PER-CRLS	The Specialist	Knows the goal, ready for higher levels, builds over time, works alone	12 months	Focus	Light
PER-CRLM	The Protégé	Knows the goal, ready for higher levels, builds over time, learns best with guidance	12 months	Focus	Guided
PER-EFNS	The Self-Starter	Still exploring, starts from basics, needs results soon, works alone	6 months	Explore	Light
PER-EFNM	The Newcomer	Still exploring, starts from basics, needs results soon, learns best with guidance	6 months	Explore	Guided
PER-EFLS	The Explorer	Still exploring, starts from basics, builds over time, works alone	12 months	Explore	Light

PER-EFLM	The Seeker	Still exploring, starts from basics, builds over time, learns best with guidance	6 months first, then extend	Explore	Guided
PER-ERNS	The Adapter	Still exploring, ready for higher levels, needs results soon, works alone	6 months	Explore	Light
PER-ERNM	The Pivoter	Still exploring, ready for higher levels, needs results soon, learns best with guidance	6 months	Explore	Guided
PER-ERLS	The Experimenter	Still exploring, ready for higher levels, builds over time, works alone	12 months	Explore	Light
PER-ERLM	The Wayfinder	Still exploring, ready for higher levels, builds over time, learns best with guidance	12 months	Explore	Guided

Preset or mode	What it sets
Focus	Goal ranking favours the learner's chosen goals; course scoring favours gap closure
Explore	Goal ranking favours interests and field of study; course scoring favours courses useful across all top goal areas

Light	Platform check-in every 2 weeks; milestones per module
Guided	Platform check-in every week; milestones per lesson; priority for mentor sessions

Re-evaluation and validation

The persona is recomputed at the end of every term. Moving from Exploring to Clear, or Foundation to Ready, is shown to the learner as progress.

Check (proposed targets)	Method	Target
Spread	Share of learners per persona	None above 20%, none empty
Independence	Correlation between any two axis scores	Below 0.4
Stability	Retake after 2 to 4 weeks on a sample	75% same persona; 85% same pole per axis
Mentor agreement	Mentors place a sample blind	70% agreement per axis
Learner recognition	"Does this sound like you?"	70% yes or partly

13. Recommendation engine

Five steps: rank goal areas, compute skill demand, measure gaps, score eligible courses, return scores with reasons. Every step is plain arithmetic on config values, so any result can be checked by hand.

Step 1. Rank career goal areas

$$T(g) = \alpha \cdot Asp(g) + \beta \cdot Fit(g) + \gamma \cdot Field(g) + \delta \cdot Mkt(g)$$

Term	Meaning	Values
Asp(g)	Learner's choice (D1)	1.0 first chosen area; 0.8 second chosen area; else 0
Fit(g)	Interest fit (Part B)	0.7 × interest value of the area's primary orientation + 0.3 × interest value of its secondary. Interest value = (scale score − 1) ÷ 4. 0 if Part B is flagged

Field(g)	Current study (A9)	1.0 direct FEEDS; 0.5 indirect; else 0
Mkt(g)	Market outlook	(outlook − 1) ÷ 4

Preset	α	β	γ	δ	Weights π for the top 3 areas
Focus (Clear)	0.50	0.20	0.20	0.10	0.60, 0.25, 0.15
Explore (Exploring)	0.10	0.45	0.30	0.15	0.34, 0.33, 0.33

If Part B is flagged, set β to 0 and divide the other three weights by their sum. Keep the top 3 areas.

Step 2. Skill demand and required level

$$D(s) = \sum_{g} \pi(g) \cdot imp(g,s) \qquad L_{req}(s) = \frac{\sum_{g} \pi(g) \cdot req(g,s)}{\sum_{g} \pi(g)}$$

imp is 1.0 for core and 0.5 for supporting. Lreq sums run only over areas that list the skill. Round Lreq to the nearest 0.5.

Step 3. Skill gap

$$Gap(s) = \max\left(0,\; L_{req}(s) - L_{now}(s)\right)$$

Sorted by D(s) × Gap(s), this is the learner's "skills to build" list.

Step 4. Score eligible courses

Eligibility. Every hard prerequisite met (completed, or placement at or above entry level), other entry rules met, not already completed.

Alignment (A). Gain counts only useful levels: up to the required level, above the current level.

$$Gain(c,s) = \max\left(0,\; \min\left(L_{out}(c,s),\, L_{req}(s)\right) - L_{now}(s)\right)$$

$$A_{raw}(c) = \sum_{s} D(s) \cdot Gain(c,s) \qquad A(c) = \frac{A_{raw}(c)}{\max_{c'\,\in\,eligible} A_{raw}(c')}$$

Unlock value (U). For each course c' that has c as a hard prerequisite, compute its raw alignment on the current state as if eligible.

$$U(c) = 0.5 \cdot \min\left(1,\; \frac{\max_{c'} A_{raw}(c')}{\max_{c''\,\in\,eligible} A_{raw}(c'')}\right)$$

Breadth (B). Share of the 3 target areas for which the course closes at least one gap on a skill that area lists: 0, 0.33, 0.67 or 1.

Format fit (F). Weekly load = course total hours ÷ 26. Available = midpoint of A14 band (under 3 ⟶ 2; 3 to 5 ⟶ 4; 6 to 8 ⟶ 7; over 8 ⟶ 10). F = 1 if load ≤ available; 0.5 if load ≤ 1.5 × available; else 0. Halve F if connectivity is "mostly offline" and the course has no offline mode.

Final score.

Score(c) = Elig(c) \cdot P_{soft}(c) \cdot \left(w_A A + w_U U + w_B B + w_F F\right)
Elig is 1 or 0. P_soft is 0.8 if a soft prerequisite is missing, else 1.

Preset	wA	wU	wB	wF
Focus	0.60	0.20	0.10	0.10
Explore	0.35	0.20	0.35	0.10

Ties and floors. Ties: higher A, then lower entry level, then course code alphabetically. A course under 0.20, or with A = 0, is never recommended.

Step 5. Reasons

Each recommended course carries: the top 3 skills it builds for this learner (by D × Gain) with from and to levels; the goal areas it serves; the course it unlocks; warnings (soft prerequisite missing, heavy load). The platform renders these with fixed sentence templates (Section 15).

Config keys (v1.0.0)

Key	Default
goal.weights.focus	α 0.50, β 0.20, γ 0.20, δ 0.10
goal.weights.explore	α 0.10, β 0.45, γ 0.30, δ 0.15
goal.pi.focus / goal.pi.explore	0.60, 0.25, 0.15 / 0.34, 0.33, 0.33
goal.asp	first 1.0, second 0.8
goal.fit_split	primary 0.7, secondary 0.3
importance	core 1.0, supporting 0.5
course.weights.focus	A 0.60, U 0.20, B 0.10, F 0.10
course.weights.explore	A 0.35, U 0.20, B 0.35, F 0.10
course.unlock_factor / soft_prereq_penalty / min_score	0.5 / 0.8 / 0.20
term.weeks	26
persona.axis_weights	as in Section 12

persona.borderline / tiebreak_weight / max_tiebreaks	0.15 / 0.25 / 2
assessment.quality	LP-Q1 = 4; LP-Q2 ≤ 2; straight-line ≥ 36 of 40; speed < 120 s

14. Pathway builder

A pathway is 1 term of 6 months with 1 course, or 2 terms over 12 months with 2 courses, one per term. Term 2 is chosen only after simulating that the learner passed term 1.

Rules

1. Length comes from the persona (Section 12). The learner can switch between 6 and 12 months before term 1 starts.
2. One course per 6-month term. v1 never runs two courses in parallel.
3. Term 1 = the highest-scoring eligible course.
4. Before choosing term 2, simulate term 1 as passed: raise its skills to its output level, mark it completed, recompute eligibility, gaps and scores.
5. Term 2 = the highest-scoring course in the recomputed list.
6. If no course reaches the minimum score, the pathway is shorter. Say the catalogue gap is closed and list the remaining gaps as "build elsewhere".
7. Explore-preset learners also get one Explore option: the best course for the 4th-ranked goal area. It sits beside the pathway, not inside it.
8. A started term is locked. The engine never changes a course in progress.

Algorithm

```
build_pathway(learner, config):
    persona = match_persona(learner)                    # Section 12
    preset  = persona.direction == 'C' ? 'focus' : 'explore'
    length  = learner.override_length or persona.default_length        # 6 or 12
    goals   = top_3(rank_goal_areas(learner, preset))           # Step 1
    state   = copy(learner.skill_levels)
    done    = copy(learner.completed_courses)
    terms   = []
    for term in 1 .. (length == 12 ? 2 : 1):
    demand = skill_demand(goals, preset)                # Step 2
    scored = score_courses(state, done, demand, preset)      # Steps 3 and 4
    best   = first(scored where score >= config.min_score)
    if best is none: break                          # catalogue gap closed
    terms.append(best with reasons)                     # Step 5
    state  = raise_levels(state, best)                  # simulate completion
    done   = done + best
```

```
    explore = preset == 'explore' ? best_course_for(goal_rank_4) : none
    return pathway(persona, goals, terms, explore, config.version)
```

Re-planning triggers

Event	What the engine does
Course passed	Raise skill levels (source: course assessment). At term end, recompute the next term.
Course not passed	Offer a retake next term, or the next best course at the same level. Recompute term 2.
Goals changed (Part D re-answered)	Recompute persona and goal areas. Keep the current term; recompute future terms.
End of term	Re-run Part C for the series studied, Part B habit scales and Part D. Recompute persona and future terms.
Pathway finished	Build a new pathway from the updated state.
New course activated	Existing pathways stay. New runs include it. A learner is told only if their next unstarted term would change.

Pathway record (API response)


```
{
  "pathway_id": "PW-000123",
  "learner_id": "L-004512",
  "config_version": "v1.0.0",
  "persona": {"code": "PER-CFLS", "name": "The Climber", "confidence": "Medium"},
  "length_months": 12,
  "goal_areas": ["CG-02", "CG-11", "CG-01"],
  "terms": [
        {"term": 1, "months": "1-6", "course_id": "COM-3", "score": 0.90,
        "reasons": {"skills": ["SK-COMM-007 3 to 4", "SK-COMM-004 3 to 4"],
        "goal_areas": ["CG-02", "CG-11", "CG-01"],
        "unlocks": "COM-4", "warnings": []}},
    {"term": 2, "months": "7-12", "course_id": "AAI-1", "score": 0.90}
  ],
  "explore_option": null,
  "build_elsewhere": ["SK-THK-001 Numerical accuracy checking"]
 }
```

15. Career outcomes

Career outcomes are shown as skills built and the kinds of work those skills are used in, never as a promised job. They come from two sources only: the skills ontology (what a course builds) and the career goal ontology (where those skills are used).

What the learner sees (results screens)

Screen card	Contents	Source
Your learner type	Persona name, its one line, what it means for the plan (length and support)	Section 12
Your interests	The 4 interest orientations as a simple bar, top one named	Part B
Where your skills could take you	Top 3 goal areas, each with its description, 3 example roles and why it was picked (your choice, your interests, your studies)	Section 8
Skills to build	Top 5 gaps, each with current and needed level	Section 13
Your pathway	One card per term: course title, "designed against NSQF Level X", duration, skills it builds (from and to level), goal areas it helps, what it unlocks	Sections 9 and 14
Build elsewhere	Gaps our courses cannot close, with a plain note: "build through your degree, trade or other certificates"	Section 13

Sentence templates

- **Course:** "After [course], you are likely to be able to [terminal outcome]. These skills are used in [goal area] roles such as [role 1] and [role 2]."
- **Pathway:** "Over the next [6 or 12] months, this plan builds [skill 1], [skill 2] and [skill 3] toward [goal area]."
- **Exploring learner:** "These courses help in all three directions you are considering, so you can keep your options open while you decide."
- **Goal and interest differ:** "You chose [goal area]. Your interests lean towards [orientation] work, which also fits [other goal area]. Both are in your plan."

Wording rules

Allowed	Not allowed
"likely to be able to", "may suit", "is used in roles such as"	"will get you a job", "guaranteed", "placement"
"designed against NSQF Level 4 descriptors"	"NSQF Level 4 certified", "NCVET approved" (until confirmed)
"skills employers look for in these roles"	"job-ready", "industry certified" (unless a named certificate exists)
Persona names and interest orientations	Habit scale scores shown to learners as labels ("low discipline")

No salary figures appear in v1. Mentors see all scale scores and the confidence check; learners see the persona, interests and plan.

16. Database schema

Postgres, 33 tables in seven groups. Reference and assessment-bank tables are loaded from versioned seed JSON in v1. Learner and engine tables are written by the services.

Tables

Group	Table	Key columns
Skills	skill_family	family_id PK, name, status
Skills	skill_cluster	cluster_id PK, family_id FK, name
Skills	skill	skill_id PK, cluster_id FK, name, definition, evidence_type, transferable, status, version
Skills	skill_level_descriptor	skill_id FK, level_band, descriptor
Skills	skill_alias	skill_id FK, alias
Skills	skill_external_ref	skill_id FK, framework, ref_code

Career goals	career_goal	goal_id PK, name, definition, primary_orientation, secondary_orientation, market_outlook, kb_track_ids, status
Career goals	goal_example_role	goal_id FK, title, nco_code
Career goals	goal_skill	goal_id FK, skill_id FK, importance, required_level, source, reviewed_on
Career goals	field_of_study	field_id PK, name, institution_type
Career goals	field_goal	field_id FK, goal_id FK, weight (1.0 or 0.5)
Courses	course_series	series_id PK, name
Courses	course	course_id PK, series_id FK, title, nsqf_level, level_out, entry_level, hours_theory, hours_practical, hours_assessment, credits, delivery_modes, offline_ok, status, version
Courses	course_prerequisite	course_id FK, requires_course_id FK, kind (hard, soft)
Courses	course_eligibility_rule	course_id FK, field, operator, value
Courses	course_module	module_id PK, course_id FK, title, hours, assessment_weight
Courses	learning_outcome	lo_id PK, module_id FK, text, type, strand, domain
Courses	course_skill	course_id FK, skill_id FK, level_out, primary
Courses	lo_skill	lo_id FK, skill_id FK, level_out

Assessment	assessment_item	item_id PK, part (A to D), scale, text_en, text_hi, options (json), key (+, R or answer), level_or_band, order_no, version
Assessment	assessment_attempt	attempt_id PK, learner_id FK, part, started_at, finished_at, flags (json)
Assessment	learner_response	attempt_id FK, item_id FK, value, answered_at, ms_taken
Persona and config	persona	persona_code PK, name, one_line, default_length, preset, support_mode
Persona and config	engine_config	version PK, json, created_at, active
Learner	learner	learner_id PK, Part A fields, created_at
Learner	learner_consent	learner_id FK, purpose, granted, guardian, at
Learner	learner_profile_score	learner_id FK, attempt_id FK, people, data, ideas, things, self_discipline, independence, motivation, goal_clarity, conf_english, conf_tech, top_orientation, interest_focus, flagged, at
Learner	learner_skill_level	learner_id FK, skill_id FK, level, source, at
Engine	persona_match, recommendation_run, pathway, pathway_term, event_log	see below

Engine output tables

```
create table persona_match (
  match_id        text primary key,
  learner_id      text references learner,
  config_version text references engine_config(version),
  axis_direction numeric, axis_start numeric, axis_goal numeric, axis_drive numeric,
```

```
  persona_code   text references persona,
  secondary_code text references persona,
  confidence     numeric,          -- mean of absolute axis scores
  tiebreaks      jsonb,            -- questions asked and answers
  created_at     timestamptz default now()
);

create table recommendation_run (
  run_id         text primary key,
  learner_id     text references learner,
  config_version text,
  inputs         jsonb,            -- full snapshot used
  goal_scores    jsonb,            -- every goal area with its four terms
  skill_demand   jsonb,            -- D, Lreq, Lnow, Gap per skill
  course_scores  jsonb,            -- A, U, B, F, Elig, Psoft, Score per course
  created_at     timestamptz default now()
);

create table pathway (
  pathway_id     text primary key,
  run_id         text references recommendation_run,
  learner_id     text references learner,
  persona_code   text references persona,
  length_months  int check (length_months in (6, 12)),
  status         text check (status in ('proposed','accepted','active','completed','replaced')),
  explore_option text references course,
  created_at     timestamptz default now()
);

create table pathway_term (
  pathway_id text references pathway,
  term_no    int check (term_no in (1, 2)),
  course_id  text references course,
  score      numeric,
  reasons    jsonb,
  locked     boolean default false,  -- true once the term starts
  starts_on  date, ends_on date,
  primary key (pathway_id, term_no)
);

create table event_log (
  event_id   text primary key,
  learner_id text references learner,
  type    text,  -- course_passed, course_failed, goals_changed, term_ended, course_added
```

```
  payload          jsonb,
  at       timestamptz default now()
);
```

17. API contract

REST over HTTPS with JSON bodies, versioned under /v1. Platform screens use learner endpoints; the Admin Console uses catalogue endpoints; the course platform uses the events endpoint. Every engine call returns a run_id so results can be replayed.

Learner endpoints (platform screens)

Method	Path	Body	Returns
POST	/v1/learners	Name, date of birth, consent per purpose	learner_id
PATCH	/v1/learners/{id}	Part A answers (one or many)	Learner
GET	/v1/assessment/items?part=B&lang=hi	none	Items for Part A, B or D in display order
POST	/v1/learners/{id}/assessment/{part}	Answers with item IDs and time taken	Stored attempt; for Part B: scale scores and flags
GET	/v1/learners/{id}/skill-check/next?series=AI	none	Next adaptive item, or done with placement
POST	/v1/learners/{id}/skill-check/answers	item_id, answer, ms_taken	Placement so far; done flag
POST	/v1/learners/{id}/persona	none (computes)	Axis scores, persona, confidence, secondary, tiebreak_needed list
POST	/v1/learners/{id}/persona/tiebreak	axis, answer	Updated persona match
POST	/v1/learners/{id}/pathway	optional length_months (6 or 12)	Pathway record (Section 14)

POST	/v1/pathways/{id}/accept	none	Pathway with status accepted
GET	/v1/learners/{id}/pathway	none	Current pathway with reasons
GET	/v1/learners/{id}/results	none	Everything the results screens show (Section 15)

Course platform endpoint

Method	Path	Body	Returns
POST	/v1/learners/{id}/events	type (course_started, course_passed, course_failed, term_ended) + payload	Event stored; re-planned pathway if one changed

Admin and mentor endpoints

Method	Path	Purpose
GET, POST, PATCH	/v1/skills	List, add (as draft) or edit skills; edits raise version
GET, POST	/v1/courses	List or add courses from the template (Section 18)
POST	/v1/courses/{id}/validate	Run onboarding checks; pass or list of failures
POST	/v1/courses/{id}/activate	Make a validated course visible to the engine
GET, PUT	/v1/goals/{id}/skills	Read or replace a goal area's skill needs
GET, POST	/v1/assessment/items	Read or add items (new items start inactive)
GET, POST	/v1/config; POST /v1/config/{version}/activate	Read, create or switch config versions
POST	/v1/simulate	Run the engine on reference learners with a

		chosen config; nothing saved
GET	/v1/mentor/learners/{id}	Mentor view: all scale scores, axis scores, goal scores, course scores, confidence check

Errors

Code	When
400	Missing or invalid field; body lists each field
409	Term is locked, or an ID already exists
422	Assessment incomplete: the response names the missing part
451	Consent missing, or guardian consent missing for a learner aged 17

18. New course onboarding template

A new course enters the engine by filling this template, passing validation and being activated. No code changes. Section E, the skills the course builds, is the most important part: it is all the engine reads to score the course.

Template

```
# Course onboarding template v1

## A. Identity
- Course ID:          [SERIES-N, e.g. AAI-5. Never reused]
- Title:              [learner-facing title]
- Series:             [existing series ID, or new series name]
- Provider:           [In-house | partner name]
- Status:             [draft]

## B. Level and credit
- NSQF level (designed against): [1 to 8, steps of 0.5]
- Output level for tagged skills: [usually the upper end of the NSQF band]
- Hours: theory [ ] / practical [ ] / assessment [ ]
- Credits (NCrF): [training hours / 30]
- Qualification status: [internal | submitted to NCVET | NQR listed]
```

C. Entry
- Entry level: [0 to 8]
- Hard prerequisites: [course IDs, or none]
- Soft prerequisites: [course IDs, or none]
- Other eligibility rules: [field, operator, value; e.g. education_stage >= Class 12]

D. Delivery
- Modes: [facilitated lab | self-paced | blended]
- Works offline: [yes | no]
- Device minimum: [shared phone | own smartphone | laptop]
- Languages: [English | Hindi | ...]

E. Skills built (required, at least 3)
| Skill ID | Output level | Primary (Y/N) | Evidence (assessment that proves it) |
| SK-XXX-000 | | | |

New skill needed? Fill Section H first.

F. Modules and learning outcomes
| Module | Hours | LO ID | Learning outcome | Type (theory/practical) | Skill IDs (max 3) |

G. Outcomes for learners
- Terminal outcome (one sentence, starts with a verb):
- Assessment summary (gates, capstone):
- Certificate issued:
- Goal areas this course mainly serves (for review only; the engine works this out from skills):

H. New skill request (only if Section E needs a skill not in the ontology)
- Proposed name / definition / family / cluster:
- Why no existing skill fits:
- Level descriptors for the bands this course covers:
- Aliases:

Validation checks (**/v1/courses/{id}/validate**)

#	Check	Fails when
1	Unique ID	Course ID already exists or was retired
2	Level consistency	Output level below entry level, or outside 0 to 8
3	Skills exist	Any skill ID is not active

4	Enough skills	Fewer than 3 skills, or none marked primary
5	Outcome coverage	A learning outcome maps to more than 3 skills, or a primary skill has no learning outcome
6	Prerequisites valid	A prerequisite does not exist, or prerequisites form a loop
7	Reachable	The course scores above 0 for at least one goal area in a test run (/v1/simulate)
8	Wording	Title or outcomes contain banned words ("guaranteed", "placement", "certified" without a named certificate)

Filled example: AAI-1


```
{
  "course_id": "AAI-1",
  "title": "AI First Steps: Use It and Check It",
  "series_id": "AAI",
  "provider": "In-house",
  "nsqf_level": 1,
  "level_out": 1,
  "hours": {"theory": 5, "practical": 10, "assessment": 1},
  "credits": 0.5,
  "entry_level": 0,
  "prerequisites": {"hard": [], "soft": []},
  "eligibility": [{"field": "age", "operator": ">=", "value": 17}],
  "delivery": {"modes": ["facilitated_lab"], "offline_ok": true, "device_min": "shared_phone",
"languages": ["en", "hi"]},
  "skills": [
   {"skill_id": "SK-DAI-001", "level_out": 1, "primary": true,  "evidence": "Lab task: set up assistant, first
real request"},
        {"skill_id": "SK-DAI-002", "level_out": 1, "primary": true,  "evidence": "Lab task: framed request
with context, role, task, format"},
        {"skill_id": "SK-DAI-003", "level_out": 1, "primary": false, "evidence": "Practice task: five
refinement moves"},
        {"skill_id": "SK-DAI-004", "level_out": 1, "primary": false, "evidence": "Section mastery check"},
        {"skill_id": "SK-ETH-001", "level_out": 1, "primary": true,  "evidence": "Gate: find planted
fabrications in a timed task"},
        {"skill_id": "SK-ETH-002", "level_out": 1, "primary": false, "evidence": "Checkpoint on an unseen
```

task"}
  ],
  "terminal_outcome": "Use an approved AI tool on a routine task and check the result before using it.",
  "status": "draft"
 }

19. Governance and versioning

The top of every ontology is locked and the bottom grows through a fixed change process. Every change is versioned, so any past recommendation can be replayed exactly.

What can change, and how

Item	Rule	Who approves
8 skill families	Locked; change only through a design review	Pod lead
Skill clusters	Rare; only when a whole area of skill is missing	Pod lead + content governance
Skills	Add through Section H of the course template; edits raise version; retire, never delete	Content governance
12 career goal areas, 4 interest orientations	Locked in v1	Pod lead
Example roles	Grow freely	Pathways research
Goal-area skill needs, market outlook	Reviewed once a year, or when the catalogue changes	Pathways research + course teams
Courses	Template, validation, activation (Section 18)	Content governance
Assessment items	Any wording change creates a new item version; new items run as unscored trials before they count	Content governance
4 persona axes, 16 personas	Locked in v1; weights and thresholds tuned through config	Pod lead
Engine config	New version for any change; tested with /v1/simulate before activation	Product manager

Versioning

Change	Version step	Example
A weight or threshold	Patch: v1.0.0 → v1.0.1	Borderline band 0.15 → 0.12
A new course, skill, goal mapping or item	Minor: v1.0.1 → v1.1.0	BOT-1 activated
A new formula term, axis parameter or scale	Major: v1.1.0 → v2.0.0	Local norms for Part B

Every recommendation_run stores the config version, the reference data version and the item bank version. A replay with the same versions gives the same result.

Change checklist

- ☐ Change recorded with reason, author and date
- ☐ Seed JSON updated and migration written
- ☐ /v1/simulate run on the 20 reference learners; differences reviewed
- ☐ No locked term changed
- ☐ Learner-facing wording checked against Section 15

20. Worked example

Learner A, a second-year B.Com student in Bihar, comes out as PER-CFLS (The Climber) and gets a 12-month pathway: COM-3 in months 1 to 6, then AAI-1 in months 7 to 12. Every number follows Sections 10, 12 and 13. Market outlooks are illustrative.

Assessment answers

Part	Answers
A	Age 19, Bihar; college; UG year 2; about 1 year left; B.Com; no certificates; no work experience; own smartphone and college lab; reliable internet; 3 to 5 hours a week
B	Interests: Data 4.25, People 3.50, Ideas 3.00, Things 2.25. Self-discipline 3.80; Learning independence 4.00; Learning motivation 3.75; Goal clarity 3.00; Confidence English 3.00; Confidence Tech and AI 2.67. All quality checks passed
C	AI placement 0; Communication placement 3; finished in one sitting at normal pace

D	Chose one goal area: CG-02 Finance, Accounts & Banking. Main goal: job in 1 to 2 years. Earning need: 1 to 2 years. Within state

Persona

Axis	Parameter values × weights	First score	Tie-breaker	Final	Pole
Direction	clarity 0 × 0.35; one area +1 × 0.30; focus 0.75 → +0.5 × 0.20; agreement +1 × 0.15	0.55	none	0.55	C Clear
Starting Point	AI −1 × 0.35; Comm 0 × 0.35; education 0 × 0.20; device +1 × 0.10	−0.25	none	−0.25	F Foundation
Goal Horizon	goal 0 × 0.50; earning +0.3 × 0.30; time left 0 × 0.20	0.09 (borderline)	"Better qualified": +1 × 0.25	(0.09 + 0.25) ÷ 1.25 = 0.27	L Long-term
Learning Drive	discipline +0.40 × 0.30; hours −0.3 × 0.20; motivation +0.375 × 0.15; independence +0.5 × 0.15; diligence +1 × 0.10; confidence −0.08 × 0.10	0.28	none	0.28	S Self-driven

Result: **PER-CFLS, The Climber**. Confidence (0.55 + 0.25 + 0.27 + 0.28) ÷ 4 = 0.34, Medium. Secondary persona: PER-CRLS (Starting Point is the weakest axis). Default: 12 months, Focus preset, Light support.

Goal-area ranking (Focus: α 0.50, β 0.20, γ 0.20, δ 0.10)

Goal area	Asp	Fit	Field	Mkt	T(g)

CG-02 Finance, Accounts & Banking	1.0	0.76	1.0	0.75	**0.93**
CG-11 Data & Analytics	0	0.72	0.5	1.00	**0.34**
CG-01 Office & Administration	0	0.76	0.5	0.50	**0.30**
CG-03 Sales, Marketing & Customer Service	0	0.59	0	0.75	0.19
CG-09 IT Support & Digital Operations	0	0.46	0	0.75	0.17

Fit for CG-02 = 0.7 × Data (0.81) + 0.3 × People (0.63) = 0.76. Targets and weights π: CG-02 0.60, CG-11 0.25, CG-01 0.15.

Skill demand and gap

Skill	D(s)	Lreq	Lnow	Gap
SK-THK-001 Numerical accuracy checking	0.925	4	0	4
SK-COMM-007 Report and business writing	0.875	4.5	3	1.5
SK-ETH-001 Output verification	0.850	4	0	4
SK-COMM-004 Reading work documents	0.750	4	3	1
SK-DAI-002 Prompt framing	0.700	3	0	3
SK-COMM-009 Workplace conversation	0.375	4	3	1

SK-DAI-006 Task decomposition	0.250	3	0	3
SK-COMM-024 Email and chat	0.150	4	3	1
SK-ETH-004 Data privacy and masking	0.150	4	0	4

Term 1 scoring

Eligible: AAI-1, COM-1, COM-2, COM-3. COM-1 and COM-2 build only to levels 2 and 3, so their gain is 0 and they drop out.

Course	A_raw	A	U	B	F	Score
COM-3	0.875 × 1 + 0.75 × 1 = 1.625	1.00	0.5	1.0	1.0	**0.90**
AAI-1	0.85 × 1 + 0.70 × 1 = 1.550	0.95	0.5	1.0	1.0	0.87

Term 1 = COM-3. Simulated completion raises its skills to level 4.

Term 2 scoring

Course	A_raw	A	U	B	F	Score
AAI-1	1.550	1.00	0.5	1.0	1.0	**0.90**
COM-4	0.875 × 0.5 = 0.438	0.28	0	1.0	1.0	0.37

Term 2 = AAI-1.

What Learner A sees

- **Learner type:** The Climber: you know where you want to go, you are starting from the basics, and you are building over time on your own.
- **Interests:** strongest in Data work, then People work.
- **Where your skills could take you:** Finance, Accounts & Banking (your choice and your B.Com), Data & Analytics and Office & Administration (your interests and studies).
- **Months 1 to 6, COM-3 Supervisory Communication Skills:** builds report writing and reading work documents from level 3 to 4. Used in all three goal areas. Unlocks COM-4.
- **Months 7 to 12, AAI-1 AI First Steps:** builds output verification and prompt framing from level 0 to 1. Unlocks AAI-2.

- **Build elsewhere:** numerical accuracy checking to level 4, through your B.Com coursework.

The next pathway, run after month 12, would most likely start with AAI-2.

21. Measurement, test quality and tuning

Because the assessment is new, it must prove it measures what it claims before its results are trusted. v1 is tuned by people reading results, not by a model.

Pilot the assessment before launch

Run the full assessment with about 100 learners from 2 or 3 partner institutions (proposed).

Check	Method	Target (proposed)
Scale reliability	Cronbach's alpha per Part B scale	0.70 or above; 0.60 accepted for 3-item scales
Item quality	Item-total correlation	0.30 or above; rewrite or drop weaker items
Skill-check difficulty	Share correct per item	Between 0.30 and 0.85 at its level
Interest spread	Learners per top orientation	No orientation above 50%
Completion time	Median total time	25 to 35 minutes
Hindi parity	Same learners answer both languages on a sample	Scale scores within 0.3 on average
Understanding	Short think-aloud with 10 learners	Every item understood without help

Reference learners (test set)

Build 20 hand-made learner profiles before launch: one for each of the 16 personas, plus 4 edge cases (Part B flagged, "not sure" on all goals, learner aged 17, learner with no eligible course that has any gain). The team writes down the expected persona, goal areas, gaps and pathway for each. Every config change runs all 20 through /v1/simulate. These 20 are also the automated test suite for the AI build tool.

Live metrics

Metric	Definition	Reviewed
Assessment completion	Learners finishing all four parts ÷ starters	Weekly
Quality flag rate	Part B attempts flagged	Weekly

Tie-breaker rate	Learners needing at least one tie-breaker	Monthly
Persona spread	Share of learners per persona	Monthly
Pathway acceptance	Accepted without change ÷ proposed	Monthly
Length override	Learners switching between 6 and 12 months	Monthly
Course completion	Completed ÷ started, by persona and course	Per term
Skill growth	End-of-term skill check vs Day 0	Per term
Habit growth	End-of-term Part B habit scales vs Day 0	Per term
Learner recognition	"Does this sound like you?" yes or partly	Monthly

Tuning rules

1. Change one config value at a time and record why.
2. If persona spread is skewed, move thresholds before axis weights.
3. If acceptance is low for one preset, review its course weights before its goal weights.
4. If a course is never recommended, check its skill tags and goal-area skill needs before changing any weight.
5. Never tune on fewer than 100 learners for a persona-level change.
6. After about 500 learners, build local norms for Part B and switch to percentile-based values (major version).

22. Assumptions and open items

v1 can be built now on the assumptions below. The open items fill seed files and test content; none blocks the engine code.

Assumptions

#	Assumption	If wrong
1	The engine runs behind the Swift AI Academy platform's own screens; no chatbot is used	Add a client; the API does not change

2	The assessment is our own Swift Pathway Assessment; no external test or career knowledge base is used	Crosswalk fields (kb_track_ids) allow a link later
3	13 launch courses: AAI-1 to AAI-4, COM-1 to COM-8, BOT-1	Update the course seed
4	The four Applied AI units become a Level 1 to 4 ladder of about 16 hours each	Update levels and hours in the course seed
5	A 12-month pathway runs its 2 courses one after the other	Allow parallel courses in v2 with a load check
6	A course builds each tagged skill to the upper end of its NSQF band	Set per-skill output levels in course_skill
7	12 goal areas are enough to describe where our learners' skills are used	Add areas through a major version
8	Learners are aged 17 to 30	Add rules and items for other ages
9	Part B uses raw scores until local norms exist	Switch to norms after about 500 learners

Open items

#	Item	Needed for	Owner (role)
1	Check whether the Career Counselling team already has a skills list; map it to the 67 IDs	Skills seed	Pathways research
2	Confirm each course's skills and output levels; map learning outcomes to skills	course_skill, lo_skill	Content teams

3	Confirm skill needs and market outlook for the 12 goal areas	goal_skill, career_goal	Pathways research + course teams
4	Complete the field of study to goal area map	field_goal	Pathways research
5	Write the remaining Communication skill-check items (3 per band) and record listening audio	Part C	Content teams
6	Translate every item into Hindi and back-translate to check meaning	Item bank	Content governance
7	Pilot the assessment with about 100 learners; fix weak items	Section 21	Pathways research
8	Set confidence-check bands from pilot data	Mentor view	Pathways research
9	Write NSQF Level 1 to 4 descriptors for the Applied AI ladder	Level wording	Content governance
10	Write the BOT-1 scope; confirm level, hours and prerequisites	BOT-1 record	Content teams
11	Confirm eligibility rules for COM-5 to COM-8	Course eligibility	Content governance
12	Design consent screens, including guardian consent	Learner Service	Product
13	Build the 20 reference learners with expected outputs	Test suite	Product + pathways research
14	Confirm the course platform can send course and term events	Events API	Engineering